Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “velocity” and “speed” to explain your model or result. Sentence starters are optional.

velocity: rate of position change, with direction

speed: how fast something moves, without direction

slope: vertical change divided by horizontal change on a graph

01 Warm-up: bring back an earlier idea

Start at 3 m, move to 10 m, then finish at 5 m.
Draw the route.
Find distance, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

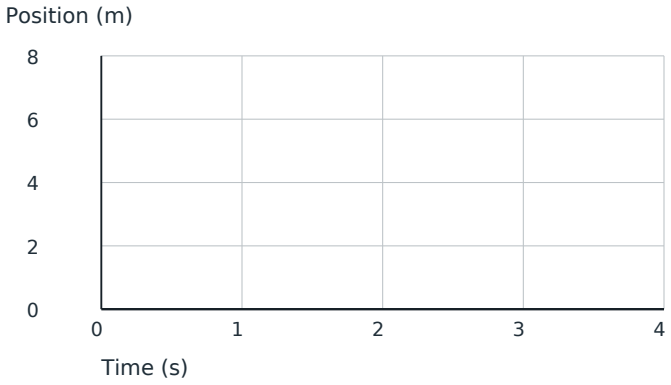
02 Worked example: explain the model

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt.
Δ means change; Δt is elapsed time. Position-time slope gives velocity.

Worked reasoning: A cart moves from 1 m to 7 m in 3 s at steady velocity. The position-time slope is (7-1)/3=2 m/s.

Your turn: Use the blank axes to plot the cart's two measured positions and connect them.
Draw a slope triangle.
Label its rise in meters and its run in seconds.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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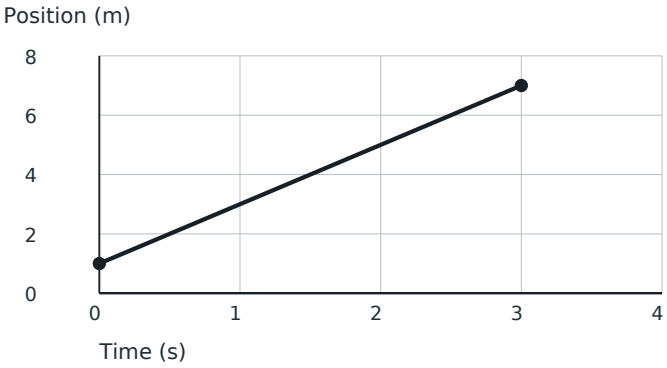
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read the motion picture

A motion sensor tracks a cart on a straight test track. Read its position at 0 s and 3 s. Which direction is it moving?
Use the changing position to explain your answer.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

04 Find the rate

Use the graph in 3. Position change = ____ - ____ = ____ m. Time = ____ s. Velocity = ____ m/s to the ____.

Sentence starter: I found … [unit]. This value means … in this situation.

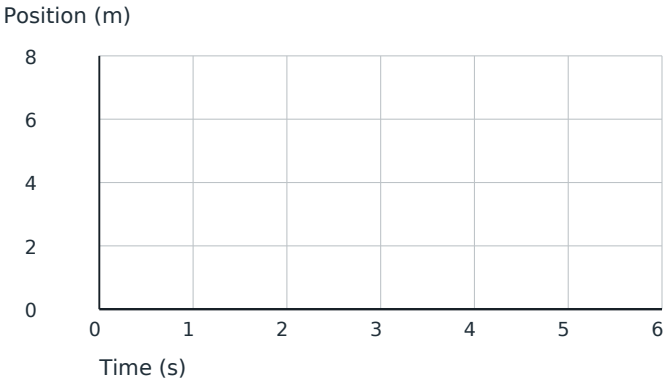
Marzano check - circle: ① ② ③

05 A pause

During another trial, the cart's position stays at 5 m from 4 s to 6 s.
Sketch that interval on the axes. Is the cart moving?
Explain what stays unchanged.

Support: The cart is ____ because its position ____.

Sentence starter: My claim is … . The diagram/data show …, so … .



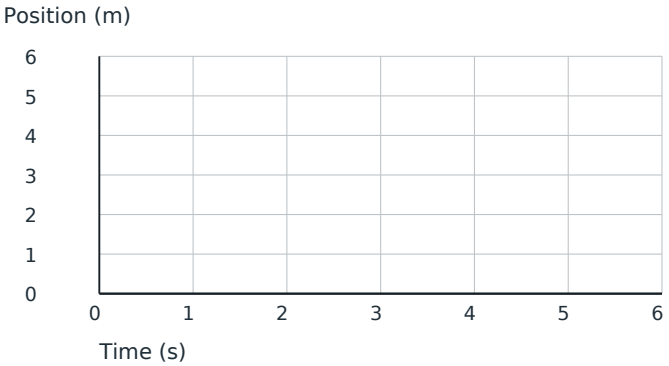
Marzano check - circle: ① ② ③

06 Draw a walk with a pause

A student walks along a marked hallway. Plot (time, position): (0 s, 1 m), (2 s, 5 m), (4 s, 5 m), (6 s, 1 m). Connect neighboring points with straight lines.
Label each interval: moving right, stopped, or moving left.

Support: Data: t = 0, 2, 4, 6 s; x = 1, 5, 5, 1 m.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Distance or displacement?

For the walk in 6, find distance traveled and displacement. Why is average velocity zero even though the person walked?

Sentence starter: My claim is … . The diagram/data show …, so … .

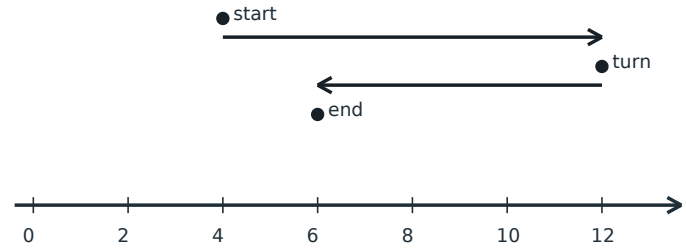
Marzano check - circle: 1 2 3

08 Review: connect an earlier idea

Start at 4 m, move to 12 m, then finish at 6 m.
Draw the route.
Find distance, displacement with direction and final position.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: I found … [unit]. This value means … in this situation.



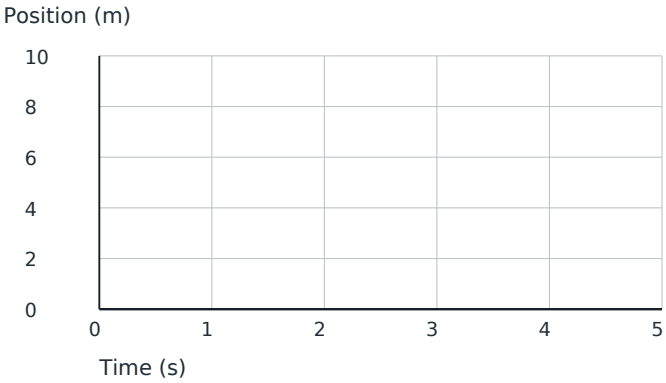
Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

A remote-control car moves steadily from the 9 m mark to the 3 m mark in 5 s.
Find its velocity, including direction, and draw its position-time graph.

Sentence starter: I found … [unit]. This value means … in this situation.

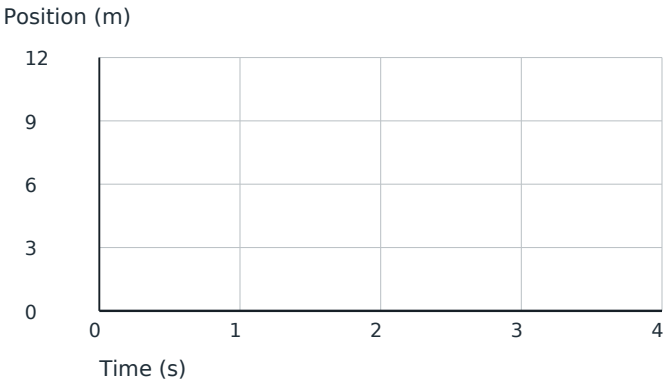


Marzano check - circle: 1 2 3

10 STRETCH: test the model

Draw two straight position-time graphs with different starting positions but the same velocity of 2 m/s.
Explain the feature they share.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3